

Engineering Reboot 2026

Week-02 Day-06 Engineering Lab Manual v4.0 - Part 1 Foundations

Mission Brief

Today's mission is to learn how software systems make decisions. Programs frequently evaluate conditions and choose between multiple execution paths. Examples include thermostats, security systems, building automation, alarm systems, and user authentication.

Learning Objectives

- Explain what a condition is
- Explain True and False evaluation
- Create IF statements
- Create IF-ELSE statements
- Create IF-ELIF-ELSE statements
- Validate multiple execution paths
- Design a simple decision tree
- Implement a complete decision program

Engineering Context

Electrical Systems: IF breaker trips THEN investigate overload.

Building Automation: IF occupancy detected THEN turn lights on.

Software Engineering: IF password valid THEN allow login.

Embedded Systems: IF motor temperature exceeds limit THEN stop motor.

Theory - What Is A Conditional?

A conditional evaluates whether something is True or False. Example: temperature > 80. The program uses the result to determine which action to perform.

Theory - IF Statement

Structure:

```
if condition:  
    action
```

Interpretation:

IF condition is true THEN perform action.

Theory - IF ELSE Statement

Structure:

```
if condition:  
    action_a  
else:  
    action_b
```

One and only one branch executes.

Theory - IF ELIF ELSE Statement

Structure:
if condition_a:
 action_a
elif condition_b:
 action_b
else:
 action_c

This allows multiple decision paths.

Repository Preparation

Open VS Code and PowerShell.
Navigate to:
C:\Projects\Engineering-Reboot-2026

Run:
git status

Expected:
On branch main
nothing to commit, working tree clean

Artifact Placement Matrix

Exercises:
Week-02\Exercises\Day-06\

Files:
high_temperature_alert.py
temperature_status.py
age_classification.py
grade_evaluation.py
decision_program.py

Notes:
Week-02\Notes\Day-06\Notes.md

Journal:
Week-02\Journal\Day-06\Journal.md

Evidence:
Week-02\Day-06\Evidence\

Deliverable Inventory

Exercise Files:
high_temperature_alert.py
temperature_status.py
age_classification.py
grade_evaluation.py
decision_program.py

Documentation:
Notes.md
Journal.md

Evidence:

Git-Status.png and exercise evidence screenshots.

Common Beginner Mistakes

1. Using '=' instead of '=='.
 2. Forgetting indentation.
 3. Incorrect ordering of if/elif/else statements.
- Always verify syntax carefully.

End of Part 1

Part 2 will contain the expanded laboratory exercises and implementation procedures.